THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET B

Chapter 4: **Carbon and its Compounds** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The valency of carbon is: 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 6

2. The molecular formula of the fourth member of the alkene series is: 1

(a) C_4H_8 **(b)** C_4H_{10} **(c)** C_5H_{12} **(d)** C_5H_{10}

3. Ethanoic acid is commonly known as: 1

(a) citric acid **(b)** formic acid **(c)** lactic acid **(d)** acetic acid

4. A student states: “Ethane decolourises bromine water.” The correct response is: 1

(a) Correct
(b) Incorrect — only unsaturated compounds do
(c) Correct in sunlight only
(d) Cannot be determined

5. For pure ethanoic acid freezing in winter, Student A calls it glacial acetic acid; Student B calls it vinegar. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. The number of single bonds in ethane (C_2H_6) is: 1

(a) 8 **(b)** 6 **(c)** 5 **(d)** 7

7. Which of these is an allotrope of carbon made of 60 atoms? 1

(a) graphite **(b)** fullerene **(c)** charcoal **(d)** diamond

8. The suffix used for an aldehyde group in IUPAC naming is: 1

(a) –ol **(b)** –al **(c)** –oic acid **(d)** –one

9. Ethanol reacts with sodium to evolve: 1

(a) oxygen **(b)** carbon dioxide **(c)** methane **(d)** hydrogen

10. The gas used in welding, which burns with a hot flame with oxygen, is: 1

(a) ethyne **(b)** ethene **(c)** methane **(d)** ethane

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is meant by tetravalency of carbon? 2

12. Write the names of the next two homologues of CH_3OH. 2

13. Why is graphite a good conductor of electricity? 2

14. A student says that consumption of small amounts of methanol is harmless. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Draw the electron-dot structures of 3
(i) ethane
(ii) ethene
(iii) carbon dioxide.

16. What is an addition reaction? Explain with an example. Give one industrial application. 3

17. What is a functional group? Name the functional groups in 3
(i) CH_3CHO
(ii) CH_3COCH_3 and write the IUPAC names.

OR

What is denatured alcohol? Why is it made? Give one harmful effect of drinking alcohol. 3

18. Why are detergents better cleansing agents than soaps in hard water? Explain with reactions in words. 3

OR

Write the structural formula of 3
(i) bromopentane
(ii) butanone
(iii) hexanal.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Explain the oxidation of ethanol with an equation. 5
(ii) What are oxidising agents? Give two examples.
(iii) Distinguish combustion from oxidation.

OR

(i) Write three reactions of ethanoic acid with equations (with base, carbonate, alcohol). 5
(ii) Why is ethanoic acid called a weak acid?

20. **Case study:** Hydrocarbons P (C_3H_8), Q (C_3H_6) and R (C_3H_4) are tested. 5
(i) Classify each as alkane/alkene/alkyne.
(ii) Which ones decolourise bromine water?
(iii) Write the structure of Q.
(iv) Which one burns with the cleanest flame?

OR

(i) What are covalent compounds? 5
(ii) Why are they poor conductors of electricity and have low melting points?
(iii) Draw the electron-dot structure of H_2O and NH_3.

END OF QUESTION PAPER — **Total: 40 Marks**

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